

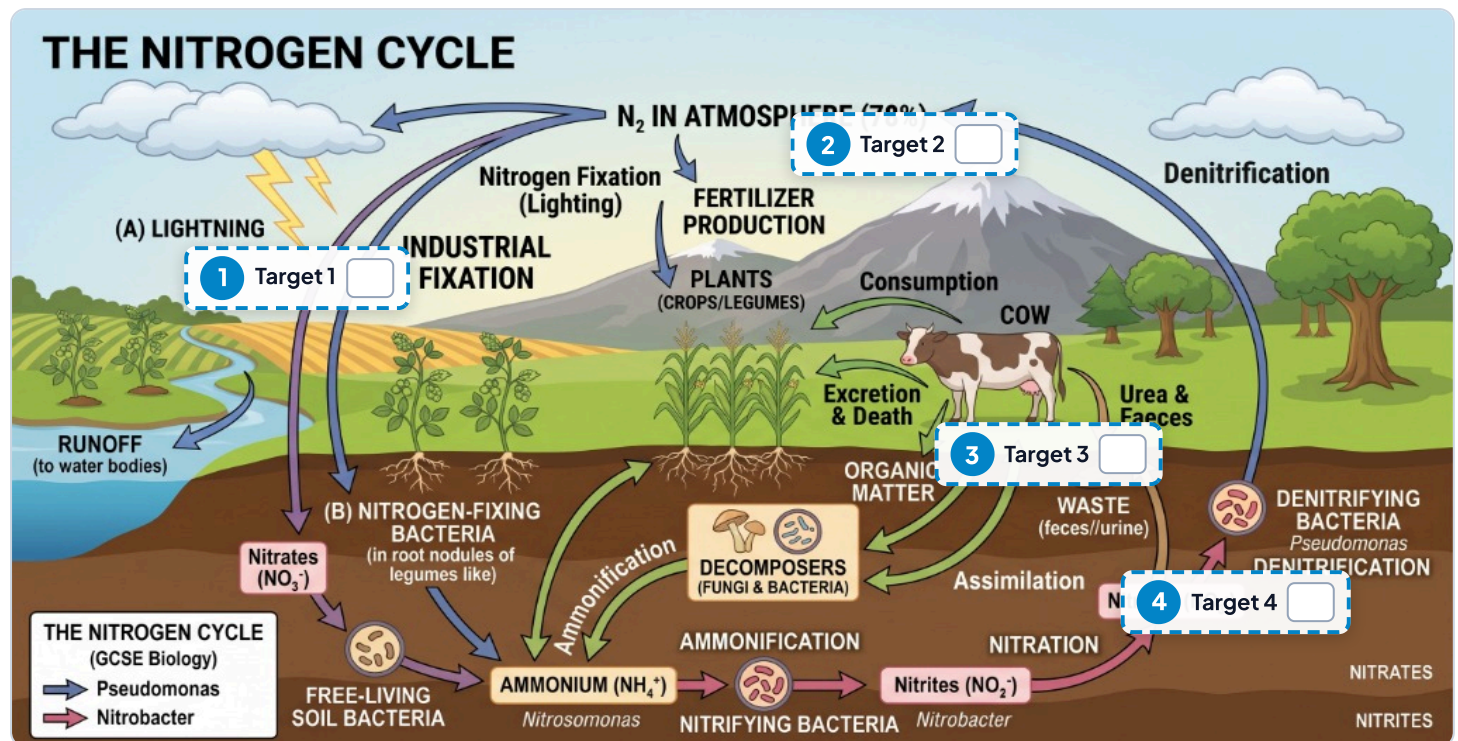
The Nitrogen Cycle: Step-by-Step Mechanism

Match the simplified mechanism cards to their correct numbered stages on the cycle

Student Handout · Printable A4

Name: _____ Date: _____ Class: _____ Score: / 4

Instructions: Study the Gemini Nitrogen Cycle diagram below. The four circular mechanism cards from Slide 7 have been simplified into Cards A, B, C, and D below. Match each Card to its correct numbered target zone ([1], [2], [3], or [4]) on the diagram. Write the matching Card Letter in the target box on the diagram, or write the Target Number on each card.



SIMPLIFIED SLIDE 7 MECHANISM CARDS (A – D)

A Fixation (Air to Soil)

Match Target:

- Nitrogen-fixing bacteria:** Found free in soil or in root nodules of legumes (peas/clover); convert unreactive N_2 gas into nitrates.
- Lightning:** Provides high electrical energy forcing atmospheric N_2 to react with oxygen, forming nitrates washed into soil.

B Assimilation & Feeding (Soil to Life)

Match Target:

- Plant Uptake:** Plant roots absorb dissolved nitrates (NO_3^-) from soil via active transport to synthesize amino acids & proteins.
- Feeding:** Animals consume plants, digesting and absorbing nitrogenous compounds to build their own animal proteins.

C Decomposition (Death to Ammonia)

Match Target:

- Waste & Mortality:** Plants and animals eventually die; animals also excrete nitrogen-rich urea and faeces during life.
- Decomposers:** Bacteria and fungi break down dead protein and waste (ammonification), releasing ammonia / ammonium into soil.

D Nitrification & Denitrification

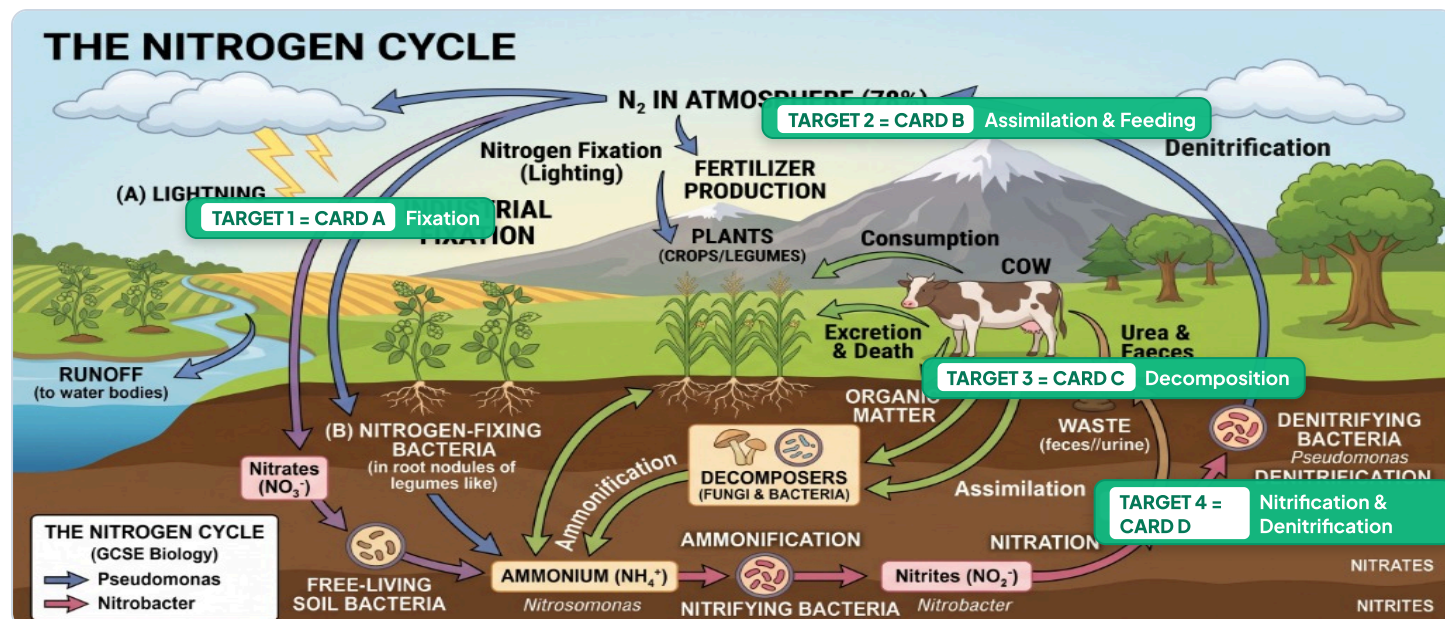
Match Target:

- Nitrification:** Nitrifying bacteria in aerobic soil oxidize toxic ammonia → nitrites (NO_2^-) → nitrates (NO_3^-) usable by plants.
- Denitrification:** In waterlogged, anaerobic soil, denitrifying bacteria convert nitrates back into N_2 gas, reducing soil fertility.

The Nitrogen Cycle Mechanism — Teacher Answer Key

Official OCR Gateway B7.1 criteria, bacterial roles, and common examiner distinctions

Mark Scheme & Misconceptions



Target	Match	Mechanism Stage & Chemistry	Microorganism / Physical Agent	Examiner Guidance & Criteria
Target 1	Card A	Nitrogen Fixation N ₂ gas in air (78%) → Nitrates (NO ₃ ⁻) / Ammonia in soil.	- Rhizobium in legume root nodules. - Free-living soil bacteria. - Lightning electrical discharge.	1 Mark: Must reference conversion of atmospheric nitrogen gas into a soluble form (nitrates/ammonium) that can enter soil.
Target 2	Card B	Assimilation & Feeding Soil nitrates → Plant protein → Animal protein.	- Crop/legume root hair cells. - Herbivorous / omnivorous animals (cow grazing).	1 Mark: Must specify plant root uptake (active transport) for protein synthesis followed by consumer feeding/digestion.
Target 3	Card C	Decomposition (Ammonification) Organic proteins / Urea → Ammonia (NH ₃ /NH ₄ ⁺).	- Saprobiontic fungi. - Decomposing bacteria. - Animal excretion (urea, faeces).	1 Mark: Decomposers break down nitrogen-rich organic waste & biomass, returning nitrogen to soil as ammonia/ammonium.
Target 4	Card D	Nitrification & Denitrification Nitrification: NH ₄ ⁺ → NO ₂ ⁻ → NO ₃ ⁻ . Denitrification: NO ₃ ⁻ → N ₂ gas.	- Nitrifying bacteria (<i>Nitrosomonas</i> & <i>Nitrobacter</i> in aerated soil). - Denitrifying bacteria (<i>Pseudomonas</i> in anaerobic soil).	1 Mark: Both processes required: recycling toxic ammonia into nitrates (aerobic) AND loss of nitrates back to atmosphere as N ₂ (anaerobic).

⚠ Critical GCSE Biology Misconceptions & Examiner Traps:

- Trap 1: Confusing Nitrogen-Fixing with Nitrifying Bacteria:** Nitrogen-fixing bacteria convert unreactive atmospheric **N₂ gas into nitrates/ammonia** (Stage 1). Nitrifying bacteria convert **soil ammonia into nitrites and nitrates** (Stage 4). Never award marks if these roles are interchanged.
- Trap 2: Why Plants Cannot Absorb Atmospheric N₂:** Nitrogen gas consists of diatomic molecules held by an extremely strong **covalent triple bond (N≡N)**. Plants lack the enzymes to break this bond; they can only absorb soluble nitrate ions (NO₃⁻) dissolved in water through root hair cells.
- Trap 3: Denitrification Condition:** Denitrifying bacteria are **anaerobic**; they thrive in waterlogged, compacted, poorly drained soils where oxygen is scarce, stripping oxygen from nitrates and releasing N₂ gas back into the atmosphere (reducing crop yield).